Ex. 1

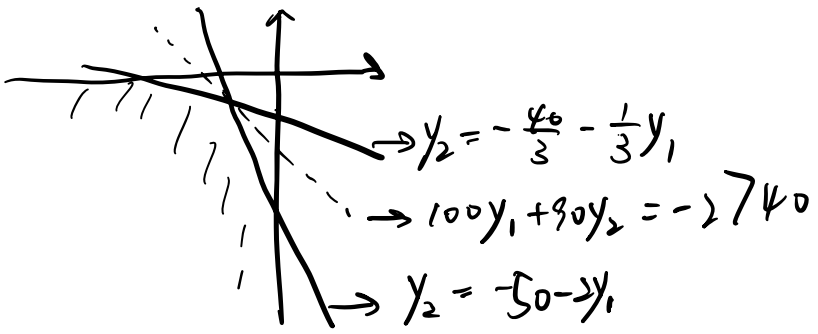
a) the primal problem is

$$\begin{aligned} \text{minimize } & -50x_1 - 40x_2 \\ \text{s.t. } & 2x_1 + x_2 \le 100 \\ & x_1 + 3x_2 \le 90 \\ & x_1, x_2 \ge 0 \end{aligned}$$

dual problem is

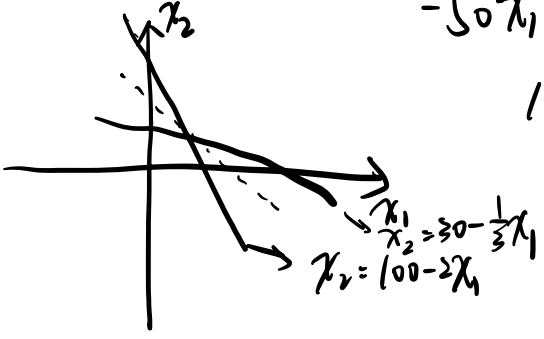
$$\begin{aligned} \text{maximize } & 100y_1 + 90y_2 \\ & 2y_1 + y_2 \le -50 \\ & y_1 + 3y_2 \le -40 \\ & y_1 \le 0, \ y_2 \le 0 \end{aligned}$$

b)



optimal value is -2740. $y^* = (-22, -6)$

c) for the primal problem



$$-50x_1 - 40x_2 \ge -2740$$

$$100y_1 + 90y_2 \le -2740$$

$\therefore \ c^T x \ge b^T y$

$\therefore$ Weak Duality theorem holds

E4.2

a) primal : min $4x_1 + 5x_2$

s.t. $x_1 + x_2 \geq 4$

$2x_1 + 3x_2 \geq 7$

$x_1, x_2 \geq 0$

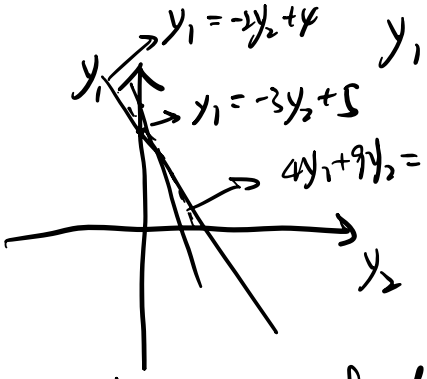
dual : maximize $4y_1 + 7y_2$

$y_1 + 2y_2 \leq 4$

$y_1 + 3y_2 \leq 5$

$y_1 \geq 0, y_2 \geq 0$

b)



∴ the opt. of dual problem is 17

$y^* = (2, 1)$

c) $c^T x^* = 12 + 5 = 17 = b^T y^*$

∴ the objective values are equal just as Strong Duality Theorem

E4.3

a) let the weight of F_1 and F_2 purchased be x_1, x_2 respectively.

the primal problem is

$$\begin{aligned} \text{minimize} \quad & 5x_1 + 4x_2 \\ \text{s.t.} \quad & 3x_1 + x_2 \geq 3 \\ & 2x_1 + 4x_2 \geq 4 \\ & x_1, x_2 \geq 0 \end{aligned}$$

b) dual problem

$$\begin{aligned} \text{maximize} \quad & 30y_1 + 40y_2 \\ & 3y_1 + 2y_2 \leq 5 \\ & y_1 + 4y_2 \leq 4 \\ & y_1 \geq 0, y_2 \geq 0 \end{aligned}$$

y_1 and y_2 can be interpreted as unit value of A and B if 3 A and 2 B's value is larger than 5, the farmer should not buy F_1, so do F_2. This helps the farmer understand the value of each nutrient in terms of cost.

c)

```
>> MAT3007_123090149_ASS3_P3
x1: 8.00
x2: 6.00
Optimal for primal: 64.00
y1: 1.20
y2: 0.70
Optimal for dual: 64.00
```

x^* and y^* are feasible, and the optimals are equal. Strong D.T. holds.

E4.4

a) let the cost matrix be $C=(c_{ij})_{2\times 3}=\begin{bmatrix} 6 & 8 & 10 \\ 7 & 5 & 9 \end{bmatrix}$

the weight transported matrix be $X=(x_{ij})_{2\times 3}$

minimize $6x_{11}+8x_{12}+10x_{13}+7x_{21}+5x_{22}+9x_{23}$

s.t.
$$\begin{aligned} x_{11}+x_{12}+x_{13} &\le 500 \\ x_{21}+x_{22}+x_{23} &\le 600 \\ x_{11}+x_{21} &\ge 300 \\ x_{12}+x_{22} &\ge 400 \\ x_{13}+x_{23} &\ge 200 \\ X &\ge 0 \end{aligned}$$

b) dual problem

$$\text{maximize } 500y_1 + 400y_2 + 300y_3 + 600y_4 + 200y_5$$

$$\text{s.t } y_1 + y_3 \le 6$$
$$y_1 + y_4 \le 8$$
$$y_1 + y_5 \le 10$$
$$y_2 + y_3 \le 7$$
$$y_2 + y_4 \le 5$$
$$y_2 + y_5 \le 9$$
$$y_1, y_2 \le 0 \quad y_3, y_4, y_5 \ge 0$$

c)

```
>> MAT3007_123090149_ASS4_P4
x11: 300.00
x12: 0.00
x13: 200.00
x21: 0.00
x22: 400.00
x23: 0.00
Optimal of primal: 5800.00
y1: -25.55
y2: -27.66
y3: 31.55
y4: 32.66
y5: 35.55
Optimal of dual: 5800.00
```

x^* and y^* are feasible, and the optimals are equal. Strong D.T. holds.

E4.5

a) maxmize 0

s.t $Ax = b$

$x \geq 0$

b) (1) according to Weak D.T. if x and y are feasible solutions,

$b^T y \geq c^T x = 0$,

however, $b^T y < c^T x$, and y is feasible, then x is infeasible

(2) $b^T y \geq 0$ for all feasible y

then the primal problem is bounded and feasible, then the dual problem is also feasible, so there must exist x such that $Ax = b$ and $x \geq 0$.

c) when (2) is true, according to b)(1), (1) is false.

when (2) is false. $b^T y \geq 0$, according to b)(2), (1) is true.

```
cvx_begin quiet
variables x1 x2 ;
    minimize 5 * x1 + 4 * x2
    subject to
        x1 * (3) + x2 >= 30;
        x1 * (2) + x2 * (4) >= 40;
        [x1, x2] >= 0;
cvx_end

%输出

fprintf('x1: %.2f\n', x1);
fprintf('x2: %.2f\n', x2);
fprintf('Optimal for primal: %.2f\n', cvx_optval);  % cvx_optval 是最优目标值


cvx_begin quiet
variables y1 y2 ;
    minimize -30 * y1 - 40 * y2
    subject to
        y1 * (3) + y2 * 2 <= 5;
        y1 + y2 * (4) <= 4;
        [y1, y2] >= 0;
cvx_end

%输出

fprintf('y1: %.2f\n', y1);
fprintf('y2: %.2f\n', y2);
fprintf('Optimal for dual: %.2f\n', -cvx_optval);  % cvx_optval 是最优目标值
```

```
cvx_begin quiet
variables x11 x12 x13 x21 x22 x23;
    minimize 6 * x11 + 8 * x12 + 10 * x13 + 7 * x21 + 5 * x22 + 9 * x23
    subject to
        x11 + x12 + x13 <= 500;
        x21 + x22 + x23 <= 400;
        x11 + x21 >= 300;
        x12 + x22 >= 400;
        x13 + x23 >= 200;
        [x11, x12, x13, x21, x22, x23] >= 0;
cvx_end

%输出
fprintf('x11: %.2f\n', x11);
fprintf('x12: %.2f\n', x12);
fprintf('x13: %.2f\n', x13);
fprintf('x21: %.2f\n', x21);
fprintf('x22: %.2f\n', x22);
fprintf('x23: %.2f\n', x23);
fprintf('Optimal of primal: %.2f\n', cvx_optval);

cvx_begin quiet
variables y1 y2 y3 y4 y5;
    minimize -(500 * y1 + 400 * y2 + 300 * y3 + 400 * y4 + 200 * y5)
    subject to
        y1 + y3 <= 6;
        y1 + y4 <= 8;
        y1 + y5 <= 10;
        y2 + y3 <= 7;
        y2 + y4 <= 5;
        y2 + y5 <= 9;
        [y1, y2] <= 0;
        [y3, y4, y5] >= 0;
cvx_end
fprintf('y1: %.2f\n', y1);
fprintf('y2: %.2f\n', y2);
fprintf('y3: %.2f\n', y3);
fprintf('y4: %.2f\n', y4);
fprintf('y5: %.2f\n', y5);
```